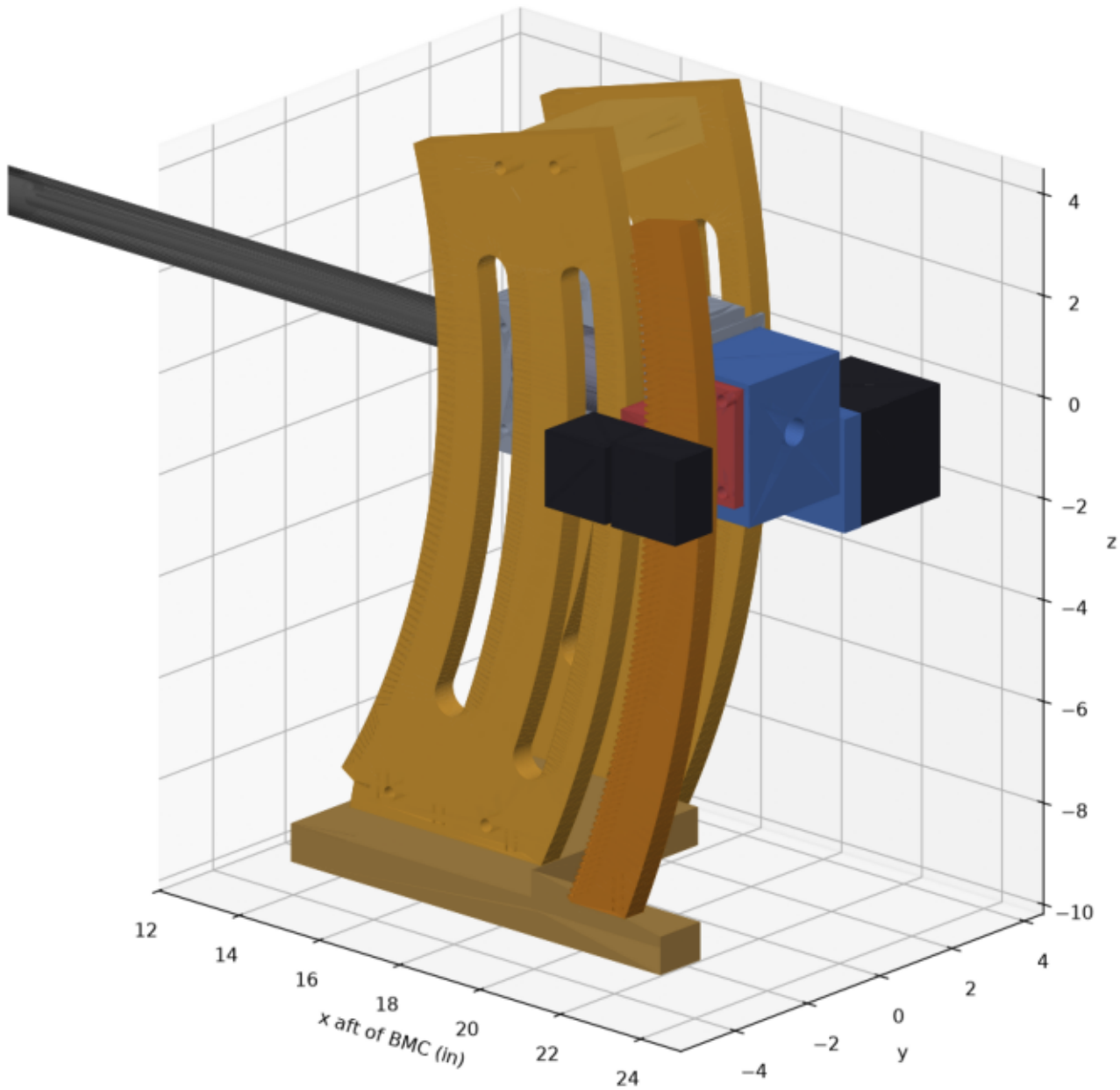


PITCH SECTOR WITH ROLL SPINDLE (CAD RENDER, NTS)

Motorized pitch sector at alpha 0: braced cheeks with arc tracks, rack plate + pinion drive (-y side), roll spindle carried on 4 cam followers



NOTES:

1. BOLT THE CHEEKS, TIE AND SPACERS TOGETHER ON THE BASE PLATE FIRST; CHECK THE CHEEK SPACING (3.000).
2. INSTALL THE CAM FOLLOWERS IN THE ROLL HOUSING; LOWER THE ROLL ASSEMBLY INTO THE ARC SLOTS.
3. SET THE RACK-PINION BACKLASH .004-.008 IN; GRAVITY KEEPS THE SAME FLANK LOADED.
4. RANGE -6..+20 DEG; PITCH RATE LIMITED TO 2 DEG/S.

UNLESS OTHERWISE SPECIFIED:

DIMENSIONS ARE IN INCHES

TOLERANCES:

.X ±.03 .XX ±.010

.XXX ±.005 .XXXX ±.0005

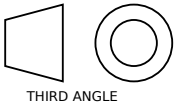
ANGLES ±0°30'

BREAK SHARP EDGES .005-.015

SURFACE FINISH 63 µin Ra MAX

INTERPRET PER ASME Y14.5-2018

UNDIMENSIONED GEOMETRY PER STEP MODEL



PARTS LIST

ITEM	PART / DWG NO.	DESCRIPTION	QTY	MATERIAL / SOURCE
1	WT-6C-401	Cheek, pitch sector	2	6061-T6
2	WT-6C-402	Rack plate, pitch	1	6061-T6
3	WT-6C-403	Plate, base, pitch	1	6061-T6
4	WT-6C-404-1	Tie, top	1	6061-T6
5	WT-6C-404-2/3	Spacers, fwd / aft	2	6061-T6
6	WT-6C-405	Bracket, pitch drive	1	6061-T6
7	WT-6C-406	Pinion shaft (+ stock m1.25 20T gear)	1	4140 + bought gear
8	-	Bearing 608-2RS	2	bought
9	-	Circlip DIN 471-8	1	bought
10	-	Cam follower 3/4 in, 3/8-24 stud	4	bought
11	-	Worm gearmotor 40:1 + NEMA 17	1	bought
12	WT-6C-200	Roll spindle assy	1	see WT-6C-200
13	WT-6C-100	Flight assy (balance, sting, model)	1	see WT-6C-100
14	-	SHCS 1/4-20 (cheeks, tie, spacers, rack)	14	bought
15	-	SHCS M6 (drive bracket)	4	bought
16	-	Limit switches + inclinometer	1	see Electronics

NEXT ASSY: WT-6C-500 YAW CARRIAGE ASSY

PROJECT				DRAWN / DATE			
Wind tunnel - motorized 6-component sting balance				Claude (AI-generated) 2026-10-02			
TITLE				CHECKED		APPROVED	
ASSEMBLY, MOTORIZED PITCH SECTOR							
MATERIAL				FINISH			
SEE PARTS LIST				SEE PART DRAWINGS			
DWG NO.	REV	SIZE	SCALE		WEIGHT		SHEET
WT-6C-400	A	B	NTS		-		4 OF 36